Comparing LLMs and Fine-Tuned Transformer Models for Named Entity Recognition and Relation Extraction in Knowledge Graph Construction

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***Abstract*—This study compares a fine-tuned biomedical transformer model with a few-shot Large Language Model approach for Named Entity Recognition and relation extraction from biomedical research abstracts. The experiments were carried out using the BioRED dataset, which provides labelled biomedical entities and relations and allows both approaches to be evaluated against the same gold-standard annotations.**

PubMedBERT was fine-tuned separately for Named Entity Recognition and relation extraction, while Qwen2.5-7B-Instruct was evaluated using structured few-shot prompts without task-specific fine-tuning. For Named Entity Recognition, PubMedBERT achieved a precision of 0.8500, recall of 0.8851, and F1-score of 0.8672, while Qwen achieved a precision of 0.6484, recall of 0.5445, and F1-score of 0.5768. For relation extraction, PubMedBERT achieved an accuracy of 0.7048 and weighted F1-score of 0.7030, compared with 0.5331 accuracy and 0.5431 weighted F1-score for Qwen. Qwen achieved higher macro recall for relation extraction, indicating greater sensitivity to less frequent relation classes, although this was accompanied by lower precision.

Overall, the results show that fine-tuned PubMedBERT provides more reliable performance for structured biomedical information extraction when labelled training data is available. Qwen offers greater flexibility through few-shot prompting, but its predictions require stricter validation and post-processing. The extracted entities and relations were subsequently used to construct a biomedical knowledge graph in Neo4j, demonstrating how the extraction outputs can be transformed into structured biomedical knowledge.

***Index Terms*—Large Language Models, PubMedBERT, Qwen2.5-7B-Instruct, Named Entity Recognition, Relation Extraction, BioRED, Biomedical Information Extraction, Knowledge Graphs**

I. Introduction

BIOMEDICAL research produces a large amount of scientific literature containing information about diseases, genes, chemicals, genetic variants, organisms, and the relationships between them. Much of this information is available only as unstructured text in research articles and abstracts, which makes it difficult to organise and reuse automatically. Converting this information into a structured form can support applications such as biomedical information retrieval, knowledge discovery, and knowledge graph construction.

Named Entity Recognition (NER) and relation extraction (RE) are two key tasks used to transform unstructured biomedical text into structured information. NER identifies important biomedical entities within text, while relation extraction determines how pairs of entities are related. When combined, these tasks can provide the nodes and relationships required for constructing biomedical knowledge graphs. Earlier work on scientific information extraction has shown that extracting entities and relations from research literature can support structured knowledge representation and knowledge graph construction [1], [2], [3].

Transformer-based language models have achieved strong performance in biomedical information extraction, particularly when they are pretrained on domain-specific biomedical text and subsequently fine-tuned using labelled data. Models such as BioBERT and PubMedBERT were developed specifically for biomedical natural language processing and have demonstrated the value of domain-specific pretraining [4], [5]. However, supervised fine-tuning depends on labelled training data and requires a separate training process for the target task.

Large Language Models (LLMs) provide a different approach to information extraction. Rather than requiring task-specific fine-tuning, instruction-tuned LLMs can perform extraction through carefully designed prompts and a small number of examples. This makes them attractive in settings where labelled training data or computational resources for fine-tuning may be limited. At the same time, biomedical extraction requires precise entity boundaries, predefined labels, and reliable structured outputs, which can be difficult to achieve consistently with generative models. Recent studies applying LLMs to biomedical information extraction have therefore highlighted both their potential and the need for careful evaluation and output validation [6], [7].

This study compares these two approaches using BioRED, a biomedical dataset containing manually annotated entities and relations [8]. The supervised pipeline uses PubMedBERT, which was fine-tuned separately for NER and relation extraction. The LLM-based pipeline uses Qwen2.5-7B-Instruct [9] in a few-shot setting without task-specific fine-tuning. Both approaches were evaluated against BioRED gold-standard annotations using precision, recall, and F1-score, with relation extraction additionally evaluated using accuracy and macro- and weighted-averaged metrics. Structured prompting, JSON validation, and post-processing were used to make the generative outputs suitable for quantitative evaluation.

The main objective of the study is to examine whether the flexibility of few-shot LLM-based extraction can provide performance comparable to a biomedical transformer that has been directly fine-tuned on labelled data. The comparison considers both overall extraction performance and practical issues such as output validity, class imbalance, and the ability to produce structured predictions that can be used in down-

stream knowledge graph construction. In particular, the study investigates the following questions:

- How does fine-tuned PubMedBERT compare with few-shot Qwen2.5-7B-Instruct for biomedical Named Entity Recognition?
- How do the two approaches compare for biomedical relation extraction?
- What practical strengths and limitations do the two approaches show when their outputs are used for structured biomedical knowledge representation?

By evaluating a domain-specific fine-tuned transformer and a general-purpose instruction-tuned LLM within the same biomedical setting, this study provides an empirical comparison of the trade-offs between supervised fine-tuning and few-shot prompting for biomedical information extraction.

II. Literature Review

The literature relevant to this study can be grouped into three main areas: scientific information extraction and knowledge graph construction, biomedical Named Entity Recognition and relation extraction, and the more recent use of Large Language Models for biomedical information extraction. These areas reflect the development of the project from a broader scientific information extraction problem towards a focused comparison of supervised biomedical transformers and few-shot LLM-based extraction.

A. Literature Selection Approach

The studies reviewed for this project were selected based on their relevance to scientific information extraction, biomedical Named Entity Recognition, relation extraction, domain-specific transformer models, Large Language Model-based extraction, and knowledge graph construction. Relevant papers were identified through academic sources including the ACL Anthology, PubMed and PubMed Central, the ACM Digital Library, Oxford Academic, and arXiv.

The review included both foundational and more recent work. Foundational studies were retained to explain how entity and relation extraction have been used for scientific knowledge representation, while biomedical studies were selected to support the choice of domain-specific models and the BioRED dataset. Recent LLM-based studies were included to examine the growing use of prompt-based approaches for biomedical extraction and to identify the practical issues that arise when generative models are used for structured information extraction.

Studies that did not directly contribute to these areas were excluded. The final literature therefore focuses on work that informed the dataset choice, model selection, evaluation strategy, and knowledge graph direction of the project.

B. Scientific Information Extraction and Knowledge Graph Foundations

Early work on scientific information extraction provided the initial foundation for this project. The Semantic Scholar Open Research Corpus (S2ORC) introduced by Lo et al. provides a large collection of scientific papers and structured metadata that can support information extraction and knowledge modelling at scale [1]. Although S2ORC was not used as the final evaluation dataset in this project, it helped establish the broader motivation for extracting structured information from scientific literature.

Luan et al. explored the joint identification of entities, relations, and coreference in scientific text and showed how these components can contribute to scientific knowledge graph construction [2]. Their work demonstrates that extracting entities alone is not sufficient for structured knowledge representation; the relationships between those entities are equally important.

Wadden et al. extended this direction using contextualized span representations for entity, relation, and event extraction [3]. Their approach showed the value of using contextual representations to perform multiple information extraction tasks within the same framework. These studies influenced the initial design of this project by demonstrating how unstructured scientific text can be transformed into structured entity and relation information.

S2AND, introduced by Subramanian et al., focuses on author name disambiguation rather than biomedical entity or relation extraction [10]. However, it remains relevant to the broader issue of data quality in scientific knowledge systems. The work highlights how ambiguity and incorrect representation can reduce the reliability of structured academic data, which is also relevant when considering the quality of entities and relations included in a knowledge graph.

TABLE I
Summary of Related Literature and Relevance to This Study

Theme	Key Studies	Main Contribution	Relevance to This Study
Scientific information extraction	S2ORC [1], Luan et al. [2], Wadden et al. [3], S2AND [10]	Scientific corpora, joint entity and relation extraction, contextual span representations, and reliable structured scientific data	Provides the original foundation for transforming scientific text into structured entities, relations, and knowledge representations
Biomedical NER and relation extraction	BioBERT [4], PubMedBERT [5], BioRED [8], EMBRE [11]	Domain-specific biomedical language models, labelled biomedical data, and entity-aware relation extraction	Supports the use of PubMedBERT, BioRED, and explicit entity representation for the supervised biomedical extraction pipeline
LLM-based biomedical extraction	Zhang et al. [6], Jahan et al. [7], Arsenyan et al. [12]	Prompt-based biomedical extraction, biomedical LLM evaluation, and knowledge graph construction	Supports the few-shot LLM pipeline and highlights the need to evaluate structured output validity and extraction reliability

C. Biomedical Named Entity Recognition and Relation Extraction

As the project became more focused on biomedical information extraction, domain-specific biomedical language models became more relevant. BioBERT, introduced by Lee et al., demonstrated that adapting BERT to biomedical corpora can improve performance across biomedical text-mining tasks [4]. This work provided early evidence that models pretrained on biomedical text are better suited to specialised biomedical terminology than general-purpose language models.

PubMedBERT further developed this idea by pretraining a transformer model entirely from biomedical text rather than adapting a model originally trained on a general-domain corpus [5]. The study showed that domain-specific pretraining can provide strong performance across biomedical NLP tasks. This directly influenced the choice of PubMedBERT as the supervised transformer model in the present study.

The BioRED dataset introduced by Luo et al. is central to this project because it contains manually annotated biomedical entities and relations within PubMed abstracts [8]. Unlike large unlabelled scientific corpora, BioRED provides the gold-standard annotations required for evaluating both NER and relation extraction. It therefore makes it possible to compare different extraction approaches within the same biomedical setting.

Relation extraction presents additional challenges because the model must not only recognise relevant entities but also determine the relationship between specific entity pairs. Li and Verspoor proposed EMBRE, an entity-aware masking approach for biomedical relation extraction [11]. Their work further highlights the importance of representing the target entities clearly when performing biomedical relation classification. This is particularly relevant to the present study, where explicit entity markers were used to identify the target entity pair during relation extraction.

Together, these studies support the use of a domain-specific biomedical transformer and a labelled biomedical benchmark for the supervised part of the comparison.

D. LLM-Based Biomedical Information Extraction

The growing use of Large Language Models has introduced another approach to biomedical information extraction. Unlike supervised transformer models, LLMs can perform extraction tasks through natural-language instructions and a small number of examples without requiring task-specific fine-tuning.

Zhang et al. investigated GPT-based models for biomedical relation extraction and showed that prompt-based models can be applied to structured biomedical extraction tasks [6]. Their findings support the possibility of using general-purpose LLMs for biomedical relation prediction while also showing that performance depends strongly on the prompting and evaluation setup.

Jahan et al. evaluated Large Language Models across several biomedical text-processing benchmark tasks and found that performance varied considerably between tasks and datasets [7]. This suggests that strong general language capabilities do not automatically translate into equally strong performance on specialised biomedical extraction tasks.

Arsenyan et al. explored the use of LLMs for biomedical knowledge graph construction through information extraction from biomedical text [12]. Their work is relevant to the downstream objective of this project because it demonstrates how LLM-generated entities and relations can contribute to structured biomedical knowledge representation.

These studies show why LLMs are attractive for biomedical information extraction: they can adapt to new tasks through prompting and do not require separate fine-tuning for every dataset. At the same time, their generative nature introduces additional challenges. Outputs may not always follow the required schema, entity boundaries may differ from gold annotations, and predicted relation labels may require validation before they can be used reliably.

E. Comparative Synthesis and Research Gap

The reviewed literature shows that supervised biomedical transformers and prompt-based LLMs have usually been studied under different experimental settings. Biomedical transformer research has largely focused on improving task-specific performance using labelled datasets and domain-specific pretraining, while recent LLM research has focused more on zero-shot and few-shot adaptability.

This creates an important comparison gap. There is limited work directly examining how a fine-tuned biomedical transformer and a general-purpose instruction-tuned LLM behave when they are applied to the same biomedical NER and relation extraction setting. Differences in dataset, annotation schema, evaluation procedure, and model training can otherwise make it difficult to understand whether differences in performance are caused by the model itself or by the experimental setup.

Another important issue is output reliability. A supervised classifier produces predictions within a fixed label space, whereas a generative LLM can produce outputs that do not follow the expected structure or label schema. For structured biomedical information extraction, this distinction is important because invalid entities or relations can directly affect downstream knowledge graph quality.

The present study addresses this gap by comparing fine-tuned PubMedBERT with few-shot Qwen2.5-7B-Instruct [9] using BioRED as the common biomedical benchmark [8]. Both approaches are evaluated for Named Entity Recognition and relation extraction against the same underlying gold-standard annotations. In addition to extraction performance, the study also considers practical aspects such as structured output validity, class imbalance, post-processing requirements, and the suitability of the resulting predictions for knowledge graph construction. Table I summarises the main groups of related work and their relevance to this project.

III. Dataset

A. BioRED Dataset and Data Splits

BioRED was used as the main dataset throughout the experiments. BioRED contains manually annotated PubMed

abstracts covering biomedical entities and relations, making it suitable for both NER and relation extraction evaluation [8]. The original dataset split was retained, consisting of 400 training documents, 100 development documents, and 100 test documents.

BioRED contains six biomedical entity categories: *ChemicalEntity*, *DiseaseOrPhenotypicFeature*, *GeneOrGeneProduct*, *SequenceVariant*, *OrganismTaxon*, and *CellLine*. The relation annotations contain eight relation categories: *Association*, *Bind*, *Comparison*, *Conversion*, *Cotreatment*, *Drug_Interaction*, *Negative_Correlation*, and *Positive_Correlation* [8]. For the relation classification experiments, an additional *No_Relation* category was introduced to represent entity pairs that were not connected by an annotated relation.

B. BioRED Preprocessing and Validation

The BioRED documents were reconstructed from their original title and abstract passages before being used by either model. Passage offsets were preserved during reconstruction so that the resulting document text remained aligned with the character-level entity annotations.

The training split contained 13,351 annotated entities, while the development and test splits contained 3,533 and 3,535 entities respectively. Each entity span was validated against the reconstructed document text using its recorded character offsets. All annotated entities were successfully aligned, with no offset mismatches found across the three dataset splits.

Relation annotations were also checked by resolving the entity identifiers associated with each relation. This ensured that the entity pairs used later for relation classification referred to valid annotations in the reconstructed documents.

These validation steps were carried out before model training to reduce the risk of preprocessing errors being mistaken for model errors during evaluation.

C. NER Data Preparation

For PubMedBERT, the BioRED entity annotations were converted into the BIO sequence-labelling format. Each of the six biomedical entity categories was represented by a beginning label (`B-`) and an inside label (`I-`), together with the outside label (`O`). This resulted in a total of 13 NER labels.

The PubMedBERT tokenizer was used with character-offset mappings so that the original BioRED entity spans could be aligned with token positions. Tokens overlapping an annotated entity were assigned the corresponding BIO label, while special tokens were excluded from entity labelling.

Some BioRED documents were longer than the 512-token input limit supported by PubMedBERT. The longest training document contained 722 tokens. Rather than simply truncating these documents, sentence-based chunking was used to preserve natural textual context. Consecutive sentences were grouped while keeping each resulting sequence within the model input limit.

After this process, the 400 training documents produced 415 NER chunks. The 100 development documents produced 103 chunks, and the 100 test documents produced 107 chunks. Final validation confirmed that none of the resulting sequences exceeded the maximum input length.

D. Relation Extraction Data Preparation

Relation extraction was formulated as a classification problem over a specified pair of biomedical entities. Each annotated BioRED relation was converted into one positive relation example containing the complete document context and the two target entities.

The target entities were explicitly marked within the input text so that the model could identify which entity pair was being classified. The first entity was represented using markers beginning with `@`, while the second was represented using markers beginning with `#`. The entity type was also included inside the corresponding marker structure.

In addition to the annotated positive relations, negative examples were generated from entity pairs that did not have a gold-standard relation. These examples were assigned the *No_Relation* label. An equal number of positive and negative examples was used at this stage. The resulting datasets initially contained 8,356 training examples, 2,324 development examples, and 2,326 test examples.

Although the total number of positive and negative examples was balanced, the individual positive relation classes remained strongly imbalanced. For example, common categories such as *Association* contained substantially more examples than rare categories such as *Conversion* and *Drug_Interaction*. This imbalance was considered during model training and evaluation.

E. Handling Long Relation Extraction Inputs

An additional problem appeared when the entity-marked relation inputs were tokenised. Standard truncation to 512 tokens sometimes removed one of the two target entity markers, particularly when the entities occurred far apart in a long document. Because relation classification requires both entity arguments to be present, these inputs could not be used reliably without further processing.

Validation identified 16 affected training examples, 11 development examples, and 26 test examples. These invalid examples were excluded from the final PubMedBERT relation classification dataset. This resulted in 8,340 training examples, 2,313 development examples, and 2,300 test examples.

For remaining long inputs, entity-preserving cropping was applied so that both relation arguments remained visible to the model. Where possible, a continuous context window containing both entities was retained. When the two entities were too far apart to fit naturally within a single window, contextual regions around both entities were retained while remaining within the model's sequence-length constraint.

The final relation inputs were revalidated before training to ensure that both target markers were present and that no sequence exceeded 512 tokens.

IV. Methodology

This study followed a comparative experimental methodology to evaluate two different approaches to biomedical Named

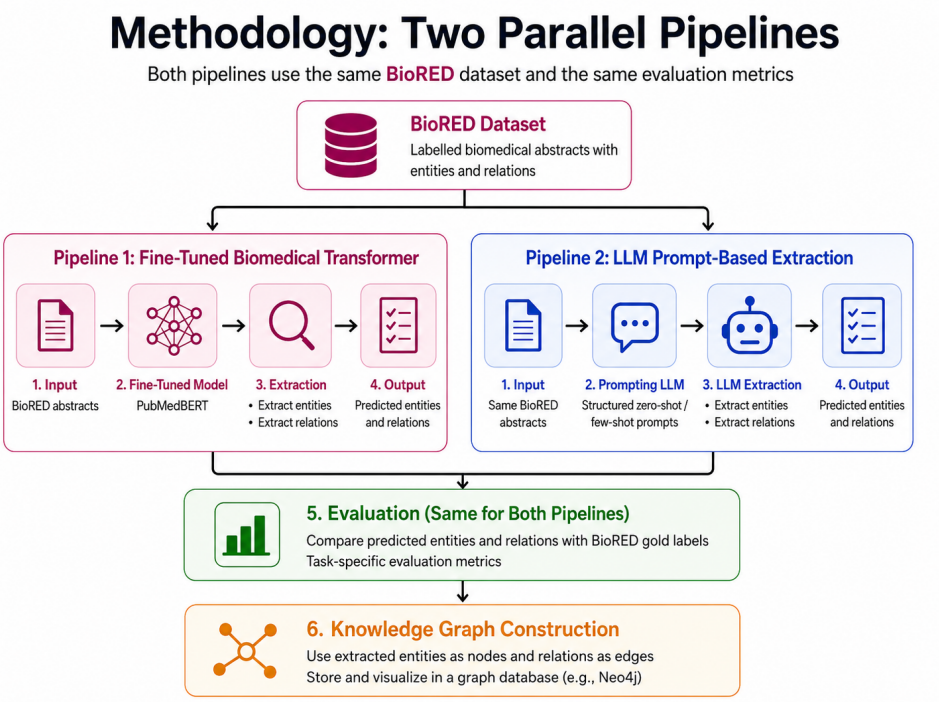


Fig. 1. Final methodology used to compare the fine-tuned PubMedBERT pipeline with the few-shot Qwen2.5-7B-Instruct pipeline for biomedical Named Entity Recognition and relation extraction.

Entity Recognition (NER) and relation extraction (RE). The first approach used PubMedBERT as a supervised biomedical transformer that was fine-tuned on labelled BioRED data, while the second used Qwen2.5-7B-Instruct [9] in a few-shot setting without task-specific fine-tuning. Both approaches were evaluated using the predefined BioRED dataset splits and the same underlying biomedical entity and relation schema.

Figure 1 provides an overview of the methodology used in this study. The workflow consists of two parallel extraction pipelines, followed by comparative evaluation and preparation of the extracted information for knowledge graph construction.

A. *PubMedBERT-Based Extraction*

For NER, PubMedBERT was configured as a token-classification model. The model learned to assign BIO labels to biomedical entities using the labelled BioRED training data.

For relation extraction, PubMedBERT was configured as a sequence-classification model with nine output classes: the eight BioRED relation categories and the additional *No_Relation* category.

B. *Qwen2.5-7B-Instruct Few-Shot Extraction*

The second extraction pipeline used Qwen2.5-7B-Instruct [9] as a general-purpose instruction-tuned Large Language Model. Unlike PubMedBERT, Qwen was not fine-tuned on the BioRED training set. Instead, the model was evaluated using few-shot prompting.

Separate prompts were developed for NER and relation extraction. In both cases, the prompts restricted the model to the BioRED label schema and required structured JSON output so that predictions could be automatically parsed and evaluated.

For the NER experiment, the prompt instructed Qwen to identify biomedical entities belonging only to the six BioRED categories and to reproduce each entity using the exact surface form appearing in the input document. The response was required to contain a single structured JSON object without additional explanation.

Unlike PubMedBERT, Qwen did not directly predict BIO tags or character offsets. It generated entity surface forms and their corresponding entity categories. A Python post-processing pipeline was therefore used to convert these outputs into a format suitable for evaluation.

First, the generated response was parsed as JSON and checked against the allowed entity labels. Each predicted surface form was then searched for in the original document. Repeated occurrences of the same surface form were expanded into separate mention-level predictions, and their exact character offsets were recovered automatically. Predictions that could not be matched to the source text were rejected during validation.

This process made it possible to compare the generative Qwen outputs with the structured BioRED annotations using strict entity matching.

For relation extraction, Qwen used the same entity-pair representation developed for the supervised relation classification pipeline. Each input therefore contained the biomedical document text together with explicit markers identifying the two target entities.

The model was instructed to select exactly one label from the predefined relation set. The required output format was:

```
{"relation": "<relation_label>"}
```

No explanatory text was requested. After generation, every output was parsed automatically as JSON and checked to determine whether the predicted relation belonged to the allowed label set. Outputs containing an invalid, unsupported, or missing relation label were treated as invalid predictions during evaluation.

C. Evaluation Strategy

NER and relation extraction were evaluated separately against the BioRED gold-standard annotations.

For NER, precision, recall, and F1-score were used as the main evaluation measures. PubMedBERT predictions were obtained from BIO sequence labels, while the Qwen-generated entity surface forms were converted into mention-level predictions through post-processing before comparison with the gold annotations.

For relation extraction, accuracy, macro precision, macro recall, macro F1, weighted precision, weighted recall, and weighted F1 were calculated. Macro-averaged metrics were included because they give equal importance to every relation category and therefore provide useful information about performance on rare classes. Weighted metrics were reported alongside them to reflect performance according to the actual class distribution.

The structural reliability of the LLM outputs was also recorded. For Qwen NER, valid JSON generation was measured. For Qwen relation extraction, both valid JSON generation and adherence to the predefined relation-label schema were checked.

Both approaches were based on the same underlying BioRED development and test splits. However, model-specific preprocessing resulted in a small difference in the number of final PubMedBERT and Qwen relation-extraction test instances. This difference is considered when interpreting the comparative results.

D. Knowledge Graph Construction

The final stage of the project used the biomedical entity information associated with the PubMedBERT relation-extraction test examples together with the predicted PubMedBERT relations to construct a biomedical knowledge graph in Neo4j. Neo4j was selected because its property-graph representation allows biomedical entities to be stored as nodes and predicted relations to be represented directly as labelled edges between those nodes.

Each extracted biomedical entity was represented as an *Entity* node. The node properties retained information such as the entity name and biomedical entity type. The relation extraction outputs were then used to create directed relationships between the corresponding source and target entity nodes. The relationship type was derived from the predicted BioRED relation category, including labels such as *Association*, *Positive_Correlation*, and *Negative_Correlation*.

The graph was constructed from the PubMedBERT extraction outputs because PubMedBERT produced the stronger overall results across the two extraction tasks, particularly for NER and for relation extraction accuracy and weighted F1. Using the more reliable extraction pipeline reduced the likelihood of introducing incorrect nodes or edges into the final structured representation.

The extracted entity and relation records were converted into a graph-compatible format before being imported into Neo4j. Entity identifiers were used to connect relation records with the corresponding source and target nodes. Model information was also retained with the relationships so that graph queries could distinguish relations originating from the PubMedBERT pipeline.

After graph construction, Cypher queries were used to inspect both the complete graph and smaller entity-centred subgraphs. This made it possible to examine how individual biomedical concepts were connected and provided a practical demonstration of how the NER and relation extraction outputs could be transformed into a queryable biomedical knowledge representation.

V. Experimental Setup

A. PubMedBERT NER Configuration

PubMedBERT was selected for the supervised extraction pipeline because it was pretrained directly on biomedical literature and has shown the benefits of domain-specific language-model pretraining for biomedical NLP [5]. The checkpoint used in this study was the PubMedBERT base uncased model trained on biomedical abstracts, available as microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract.

Before full training, a small pilot experiment was carried out to confirm that the tokenisation, BIO alignment, training, and evaluation pipeline operated correctly.

The final model was fine-tuned for three epochs using a learning rate of 2×10^{-5}, a training batch size of 4, an evaluation batch size of 4, and a weight decay of 0.01. Dynamic padding was used during batching. Model performance was evaluated on the development set after each epoch, and the checkpoint with the best development F1-score was retained for final test evaluation.

B. PubMedBERT Relation Extraction Configuration

The relation extraction model was fine-tuned for three epochs using a learning rate of 2×10^{-5} and a batch size of 4. Development performance was monitored during training, with macro F1 used as an important measure because of the strong imbalance between the individual relation categories.

The final training configuration used class-weighted cross-entropy loss to reduce the influence of the most frequent relation categories during training. Higher weights were assigned to less frequent relation categories, while frequent categories such as *Association* and *No_Relation* received lower weights. This allowed errors on minority relation classes to contribute more strongly to the training objective rather than allowing the most frequent labels to dominate the loss.

TABLE II
PUBMEDBERT NER PERFORMANCE ON THE BIORED TEST SET

Metric	Score
Precision	0.8500
Recall	0.8851
F1-score	0.8672
Accuracy	0.9684

TABLE III
QWEN2.5-7B-INSTRUCT FEW-SHOT NER PERFORMANCE

Metric	Score
Precision	0.6484
Recall	0.5445
F1-score	0.5768
Valid JSON outputs	97/100

C. Qwen2.5-7B-Instruct Configuration

Qwen2.5-7B-Instruct was used without task-specific fine-tuning. The model was loaded using 4-bit BitsAndBytes quantisation to reduce GPU memory requirements. NF4 quantisation, double quantisation, and float16 computation were used during inference. Generation was deterministic, with sampling disabled, so that the same frozen prompting setup was applied consistently throughout the evaluation.

D. Few-Shot Prompt Configuration

For the NER experiment, three documents from the BioRED training split were used as few-shot demonstrations. The selected examples contained multiple biomedical entity categories while remaining short enough to keep the complete prompt within a manageable length. The same frozen prompt configuration was then used throughout the evaluation.

For relation extraction, a frozen few-shot prompt containing nine demonstrations was used, with one example for each possible relation label. The demonstrations covered the eight BioRED relation categories as well as *No_Relation*. The same prompt was used for all relation extraction examples to ensure a consistent evaluation setup.

VI. RESULTS AND ANALYSIS

This section presents the final experimental results for PubMedBERT and Qwen2.5-7B-Instruct on biomedical Named Entity Recognition (NER) and relation extraction (RE). The two tasks are analysed separately before comparing the overall behaviour of the supervised and few-shot approaches. In addition to the main evaluation metrics, error patterns and the structural validity of the LLM outputs are considered because these factors are particularly important when the extracted information is intended for downstream knowledge graph construction.

A. PubMedBERT Named Entity Recognition Results

The fine-tuned PubMedBERT model achieved strong performance on the held-out BioRED test set. Table II presents the final NER results.

The model achieved an F1-score of 0.8672, with recall slightly higher than precision. This indicates that PubMedBERT recovered a large proportion of the annotated biomedical entities, although some additional incorrect predictions were also produced.

Performance on the development set was very similar, where the model achieved a precision of 0.8424, recall of 0.8893, and F1-score of 0.8652. The small difference between development and test performance suggests that the model generalised consistently to unseen BioRED documents rather than showing a substantial performance drop after training.

A more detailed inspection of the test predictions identified 3,129 exact entity matches, 552 false-positive predictions, and 406 false negatives. Common errors included expanded entity boundaries, merged neighbouring mentions, false-positive disease or phenotype spans, and occasional confusion between biomedical entity categories.

Among the entity categories, DiseaseOrPhenotypicFeature produced the largest number of observed errors, with 223 false positives and 154 false negatives. Chemical entities produced 143 false positives and 60 false negatives, while gene or gene-product mentions produced 90 false positives and 116 false negatives. These errors show that even with relatively strong overall performance, exact biomedical span recognition remains challenging for ambiguous or context-dependent mentions.

B. Qwen2.5-7B-Instruct Named Entity Recognition Results

Qwen2.5-7B-Instruct was evaluated on the same 100-document BioRED test split using the frozen few-shot NER prompt. Unlike PubMedBERT, Qwen was not fine-tuned using the BioRED training labels.

The final Qwen NER results are shown in Table III.

Qwen achieved an F1-score of 0.5768, which was considerably lower than the result obtained by fine-tuned PubMedBERT. Recall was particularly limited at 0.5445, indicating that a substantial proportion of gold-standard biomedical mentions were not recovered by the few-shot model.

At the same time, the structured-output mechanism worked reasonably well. Ninety-seven of the 100 test documents produced valid JSON responses. This shows that the prompt was largely successful in controlling the output format. However, structural validity did not necessarily correspond to correct biomedical extraction. Predicted entity text still had to be validated against the original document, checked against the allowed BioRED entity types, and converted into exact character offsets before evaluation.

This distinction is important for generative information extraction. Producing machine-readable output is useful for automation, but valid formatting alone does not guarantee accurate entity recognition.

C. Comparison of PubMedBERT and Qwen for NER

Table IV compares the two NER approaches.

PubMedBERT outperformed Qwen across precision, recall, and F1-score. The difference in F1-score was 0.2904, or 29.04 percentage points. The largest difference appeared in recall,

TABLE IV
COMPARISON OF PUBMEDBERT AND QWEN FOR BIOMEDICAL NER

Metric	PubMedBERT	Qwen
Precision	**0.8500**	0.6484
Recall	**0.8851**	0.5445
F1-score	**0.8672**	0.5768

TABLE V
PUBMEDBERT RELATION EXTRACTION RESULTS

Metric	Score
Accuracy	0.7048
Macro Precision	0.3870
Macro Recall	0.3990
Macro F1	0.3826
Weighted Precision	0.7100
Weighted Recall	0.7048
Weighted F1	0.7030

where PubMedBERT achieved 0.8851 compared with 0.5445 for Qwen.

A major difference between the two approaches is the amount of task-specific supervision they received. PubMedBERT was directly fine-tuned on labelled BioRED examples and learned token-level BIO patterns from the training data. Qwen received only a small number of demonstrations and had to infer the extraction task from the prompt. This made Qwen easier to adapt without additional model training, but it also provided substantially less task-specific supervision.

There was also a practical difference in the form of the predictions. PubMedBERT produced outputs within a fixed BIO label space, while Qwen generated surface forms that required JSON parsing, label validation, mention expansion, and offset recovery before they could be evaluated. The LLM pipeline was therefore more flexible, but required additional post-processing to obtain structured predictions.

The NER comparison should nevertheless be interpreted with some care. PubMedBERT was evaluated using BIO sequence-labelling predictions, whereas Qwen predictions were converted from generated surface forms and evaluated using strict mention-level matching. Both approaches were evaluated against the same BioRED test annotations using precision, recall, and F1-score, but their different output structures required different evaluation pipelines.

D. PubMedBERT Relation Extraction Results

Relation extraction produced substantially lower overall performance than the NER experiment, reflecting the additional difficulty of identifying the semantic relationship between biomedical entity pairs. The final PubMedBERT relation classifier was evaluated on 2,300 validated test examples after removing inputs where both target entity markers could not be preserved.

Table V presents the final PubMedBERT relation extraction results.

The model achieved an accuracy of 0.7048 and a weighted F1-score of 0.7030. These values indicate that the classifier performed reasonably well across the overall test distribution. However, the macro F1-score was considerably lower at 0.3826.

TABLE VI
QWEN2.5-7B-INSTRUCT FEW-SHOT RELATION EXTRACTION RESULTS

Metric	Score
Accuracy	0.5331
Macro Precision	0.3405
Macro Recall	0.6231
Macro F1	0.3873
Weighted Precision	0.5621
Weighted Recall	0.5331
Weighted F1	0.5431
Valid JSON	2326/2326
Valid relation labels	2281/2326

The difference between weighted and macro performance reflects the strong imbalance between individual relation classes. Although the relation dataset contained an approximately balanced number of positive and *No_Relation* examples before final preprocessing, the individual positive relation categories were highly uneven. For example, the training data contained 2,192 *Association* examples but only three *Conversion* examples.

Strong performance on common categories therefore contributes heavily to the weighted score, whereas macro F1 gives equal importance to every relation category. The lower macro F1 shows that the minority relation classes remained difficult despite the use of class-weighted training.

The task also required the model to distinguish between semantically similar labels such as *Association*, *Positive_Correlation*, *Negative_Correlation*, and *No_Relation*. These distinctions require more than simply identifying that two entities occur in the same biomedical context and contributed to the difficulty of the RE task.

E. Qwen2.5-7B-Instruct Relation Extraction Results

Qwen was evaluated using the frozen nine-example few-shot relation prompt. The complete prepared BioRED test relation set contained 2,326 examples.

All 2,326 model responses were successfully parsed as JSON. However, only 2,281 predictions contained a valid relation label from the predefined schema, while 45 predictions were invalid or could not be resolved to one of the allowed relation categories. These invalid outputs were counted as incorrect rather than removed from the evaluation.

The final Qwen relation extraction results are shown in Table VI.

Qwen achieved an accuracy of 0.5331 and a weighted F1-score of 0.5431. These values were below those obtained by PubMedBERT. However, Qwen achieved a macro recall of 0.6231, which was substantially higher than PubMedBERT's macro recall of 0.3990.

This difference shows that Qwen behaved differently rather than being uniformly weaker across every measure. The model was more willing to assign less frequent relation categories and therefore recovered a larger proportion of examples belonging to some minority classes. However, this higher sensitivity was accompanied by a macro precision of only 0.3405, indicating that many of these minority-class predictions were incorrect.

The effect was especially noticeable for relation types with very low support. For example, the test set contained only a

TABLE VII
COMPARISON OF PUBMEDBERT AND QWEN FOR RELATION EXTRACTION

Model	Accuracy	Macro P	Macro R	Macro F1	Weighted P	Weighted R	Weighted F1
PubMedBERT	**0.7048**	**0.3870**	0.3990	0.3826	**0.7100**	**0.7048**	**0.7030**
Qwen2.5-7B	0.5331	0.3405	**0.6231**	**0.3873**	0.5621	0.5331	0.5431

small number of examples for categories such as *Conversion* and *Drug_Interaction*. Correctly predicting even one or two examples from these classes can strongly affect macro recall, while incorrect predictions of the same labels reduce precision. Macro recall should therefore not be interpreted in isolation as evidence of stronger overall RE performance.

Error inspection also showed repeated confusion between semantically related categories. In particular, *Association*, *Positive_Correlation*, and *No_Relation* were frequently confused. This suggests that Qwen could often identify that two entities were related in some way, but had more difficulty selecting the exact relation category defined by the BioRED annotation schema.

F. Comparison of PubMedBERT and Qwen for Relation Extraction

Table VII summarises the final relation extraction comparison.

Unlike the NER comparison, the RE results were more mixed. PubMedBERT achieved higher accuracy, macro precision, weighted precision, weighted recall, and weighted F1. Its weighted F1 exceeded Qwen by 0.1599, or 15.99 percentage points. These measures suggest that PubMedBERT produced more reliable predictions across the actual distribution of the test data.

Qwen, however, achieved substantially higher macro recall and a slightly higher macro F1-score. This suggests that it distributed predictions more broadly across the relation-label space rather than concentrating mainly on the most common classes.

The difference between macro and weighted measures is particularly important when interpreting these results. Macro metrics assign equal importance to every class regardless of its frequency, while weighted metrics account for the number of examples belonging to each class. Qwen's higher macro recall therefore reflects stronger coverage of some rare relation categories, but its lower weighted F1 and accuracy show that this did not translate into stronger performance across the overall test set.

The results suggest a practical trade-off. PubMedBERT produced more reliable predictions across the overall relation dataset, whereas Qwen showed greater sensitivity to minority relation classes but at the cost of lower precision. This distinction is particularly important for knowledge graph construction, where an incorrect relation prediction may directly become an incorrect graph edge.

G. Comparison with Related Work

The experimental results can also be considered in relation to previous work on BioRED [8] and biomedical information extraction. Direct numerical comparison must be made carefully because differences in model architecture, relation formulation, input representation, and evaluation procedure can affect the reported scores.

For NER, the original BioRED study compared several supervised biomedical language models and reported a strict F1-score of 0.893 for PubMedBERT-CRF [8]. The PubMedBERT model used in the present study achieved an F1-score of 0.8672. Although slightly lower than the original benchmark, the result is broadly consistent with the BioRED findings and supports the effectiveness of biomedical domain-specific pretraining for entity recognition [5], [8].

The difference between these scores may partly reflect differences in model configuration and preprocessing. The BioRED benchmark used a CRF-based output layer, while the present implementation used a standard token-classification formulation with BIO labels. Sentence-based chunking was also used in the present study to handle BioRED documents that exceeded the PubMedBERT input length. For these reasons, the published BioRED result should be treated as useful context rather than as a direct replication.

The original BioRED experiments also showed that relation extraction remained more challenging than NER and that performance decreased when relation type information had to be predicted [8]. This general pattern is also visible in the current results, where relation extraction produced much lower macro F1 than the NER experiment. However, a direct comparison of RE scores is not appropriate because the current study formulated relation extraction as classification over explicitly marked entity pairs and introduced an additional *No_Relation* class.

More recent work has explored ways of improving relation extraction by strengthening entity-specific representations. Li and Verspoor proposed EMBRE, an entity-aware masking approach evaluated on the BioRED relation extraction challenge [11]. Their work showed that incorporating entity-specific information improved relation extraction performance over their baseline [11]. This supports the broader importance of explicitly representing the target biomedical entities during relation classification, which was also a central part of the entity-marker approach used in the present study.

The Qwen results can also be interpreted alongside broader evaluations of Large Language Models for biomedical NLP. Jahan et al. evaluated several general-purpose LLMs across multiple biomedical benchmark tasks and found considerable variation in performance depending on the task and dataset [7]. Their findings show that LLMs can perform biomedical tasks without conventional supervised fine-tuning, but specialised or fine-tuned approaches can remain stronger in settings where labelled biomedical data is available [7]. A similar pattern was observed in this study: Qwen was able to perform both NER and relation classification from a small number of demonstra-

tions, but PubMedBERT achieved substantially stronger NER performance and higher RE accuracy and weighted F1.

Overall, the comparison with related work supports the main observation of this study. Domain-specific supervised transformers remain highly effective for structured biomedical information extraction when labelled training data is available, while prompt-based LLMs provide a more flexible alternative without requiring task-specific fine-tuning. The current results suggest that this flexibility is accompanied by lower reliability for exact entity recognition and relation classification.

H. Structured Output Reliability

The Qwen experiments also allowed structural output reliability to be examined separately from predictive accuracy.

For NER, 97 of the 100 test documents produced valid JSON. For relation extraction, all 2,326 responses were valid JSON, although only 2,281 contained a valid relation label from the predefined schema. The structured prompting strategy was therefore effective at producing machine-readable outputs, particularly for the simpler single-label relation classification format.

However, structural validity and predictive correctness should not be treated as equivalent. A response may follow the requested JSON format while still containing an incorrect entity or relation prediction. The results therefore show that validation and post-processing remain necessary when generative LLM outputs are used within a structured biomedical extraction pipeline.

I. Knowledge Graph Results

The biomedical entity information associated with the PubMedBERT relation-extraction examples and the corresponding predicted relations were used to construct a biomedical knowledge graph in Neo4j. The completed knowledge graph contained 948 biomedical entity nodes and 1,126 relationships derived from the PubMedBERT extraction pipeline. The graph provides a structured representation of the biomedical information extracted by the PubMedBERT pipeline and demonstrates how the outputs of the NER and relation extraction stages can be used beyond standalone classification metrics.

Figure 2 shows a subset of the PubMedBERT-derived graph visualised in Neo4j. The displayed query contains multiple connected and disconnected biomedical components, reflecting the different entity pairs and relation predictions extracted from the source documents. The visible relation types include *Association*, *Positive_Correlation*, and *Negative_Correlation*.

A smaller entity-centred query was also used to inspect the neighbourhood of individual biomedical concepts. Figure 3 shows the subgraph associated with *SCN5A*. In this example, SCN5A is connected through *Association* relationships to several biomedical concepts, including bradycardia, tachycardia, arrhythmia, tetrodotoxin, and long QT syndrome. The returned subgraph contains six nodes and five relationships.

The knowledge graph provides a practical extension of the extraction experiments. Instead of viewing the model outputs only as independent entity and relation predictions, Neo4j makes it possible to explore them as connected biomedical information. At the same time, the graph illustrates why extraction quality is important: in a fully end-to-end pipeline, entity recognition errors could result in missing or incorrect nodes, while relation classification errors may introduce misleading edges between biomedical concepts.

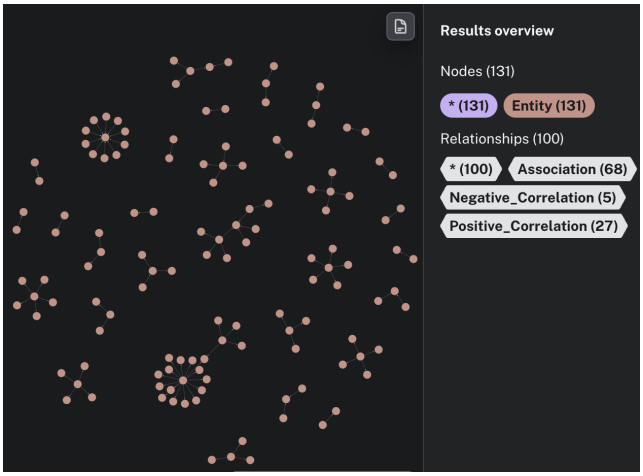


Fig. 2. PubMedBERT-derived biomedical knowledge graph in Neo4j.

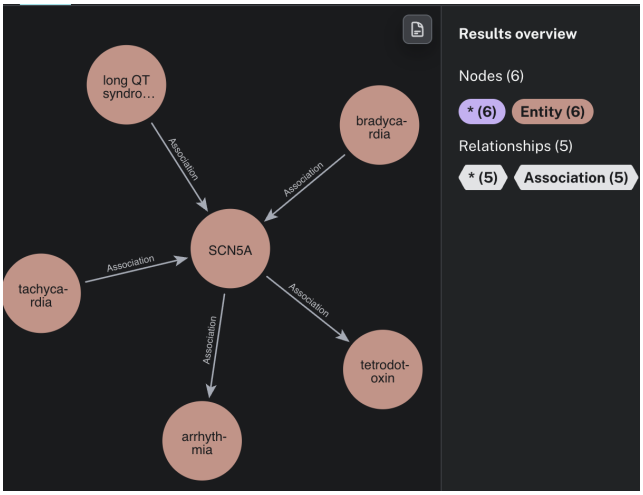


Fig. 3. SCN5A-centred subgraph in Neo4j.

J. Overall Interpretation

Across the two extraction tasks, the strongest advantage for PubMedBERT was observed in NER. Fine-tuning on labelled biomedical data resulted in substantially higher precision, recall, and F1 than few-shot Qwen. This suggests that exact biomedical entity extraction benefits strongly from domain-specific pretraining combined with task-specific supervision.

Relation extraction produced a more nuanced result. PubMedBERT remained stronger in terms of accuracy and weighted performance, while Qwen achieved higher macro recall and slightly higher macro F1. This indicates that the LLM was capable of recognising a wider spread of relation categories but was less precise when deciding which exact relation label should be assigned.

Taken together, the experiments suggest that fine-tuned PubMedBERT is better suited to settings where labelled biomedical data is available and reliable structured extraction is the main requirement. Qwen provides greater flexibility because task-specific fine-tuning is not required, but this benefit is accompanied by lower extraction performance and additional validation and post-processing requirements.

These differences are particularly relevant to downstream knowledge graph construction. NER errors may result in missing or incorrect nodes, while relation classification errors can introduce misleading edges. The stronger precision and weighted performance of PubMedBERT therefore provide an advantage when extraction reliability is prioritised. Qwen's broader minority-class sensitivity may still be useful in settings where rapid adaptation or recall is more important than precision.

K. Limitations of the Comparative Evaluation

Several limitations should be considered when interpreting the results.

First, although both NER approaches were evaluated on the same BioRED test documents, their output structures required different evaluation procedures. PubMedBERT produced BIO sequence labels, whereas Qwen generated entity surface forms that were subsequently converted into mention-level offsets. The resulting precision, recall, and F1 values therefore represent the same underlying extraction objective but were not calculated through an identical prediction pipeline.

Second, the final relation extraction test sizes differed slightly. PubMedBERT was evaluated on 2,300 validated examples after 26 inputs affected by entity-marker loss were excluded, whereas Qwen was evaluated on all 2,326 prepared test examples. The RE comparison therefore uses the same underlying BioRED test split and label schema, but not a perfectly identical set of model inputs.

Third, several BioRED relation categories contained very few examples. Performance on classes with only one or two test instances can change substantially based on a single prediction, which makes macro-averaged metrics particularly sensitive to these rare categories.

Finally, the LLM experiments evaluated one instruction-tuned model using one frozen few-shot configuration. The results therefore describe the behaviour of Qwen2.5-7B-Instruct under the selected prompting setup and should not be interpreted as representative of all Large Language Models.

VII. Conclusions

This study compared a fine-tuned biomedical transformer, PubMedBERT, with a few-shot Large Language Model, Qwen2.5-7B-Instruct, for biomedical Named Entity Recognition and relation extraction using the BioRED dataset. The aim was to examine not only which approach achieved stronger extraction performance, but also how their different strengths and limitations affect their suitability for structured biomedical information extraction and downstream knowledge graph construction.

For Named Entity Recognition, PubMedBERT clearly outperformed Qwen. PubMedBERT achieved an F1-score of 0.8672, compared with 0.5768 for Qwen. The difference was especially large in recall, showing that the fine-tuned model recovered a substantially greater proportion of the gold-standard biomedical entities. These results indicate that exact biomedical entity extraction benefits strongly from domain-specific pretraining and task-specific supervision when labelled training data is available.

The relation extraction results were more nuanced. PubMedBERT achieved higher accuracy and weighted F1, with scores of 0.7048 and 0.7030 respectively, compared with 0.5331 accuracy and 0.5431 weighted F1 for Qwen. Qwen, however, achieved higher macro recall and a slightly higher macro F1. This suggests that the few-shot LLM was more willing to predict less frequent relation categories, but this broader class coverage came with lower precision. In practical terms, PubMedBERT produced more reliable predictions across the overall test distribution, while Qwen showed greater sensitivity to minority classes.

The experiments also highlighted an important difference between supervised classification and generative extraction. PubMedBERT produced predictions directly within a fixed label space, while Qwen required structured prompting, JSON validation, label checking, and additional post-processing before its outputs could be evaluated reliably. Although Qwen demonstrated that useful biomedical extraction is possible without task-specific fine-tuning, the additional validation requirements and lower overall performance make its outputs less dependable for precise structured extraction in the current setting.

Comparison with related work showed that the PubMedBERT NER results were broadly consistent with previously reported BioRED performance, while the relation extraction findings reflected the same general difficulty associated with imbalanced biomedical relation classes. The results also align with broader biomedical LLM evaluations showing that general-purpose LLMs can perform specialised tasks through prompting, but fine-tuned biomedical models remain highly competitive when labelled domain-specific data is available.

Overall, the findings suggest that fine-tuned biomedical transformers remain the more reliable choice for structured biomedical information extraction when suitable labelled data is available. Few-shot LLMs provide an attractive alternative when flexibility, rapid adaptation, or reduced training effort is more important, but their outputs require careful validation and should not be assumed to match the precision of a task-specific biomedical model.

The extracted entities and relations provide the basis for downstream knowledge graph construction. In this context, extraction quality becomes particularly important because NER errors may create missing or incorrect nodes, while relation errors may introduce misleading edges. The final knowledge graph stage therefore serves as a practical extension of the comparison by showing how differences in extraction reliability can affect structured biomedical knowledge representation.

VIII. Use of AI

Artificial Intelligence tools were used as supporting tools during the development and writing of this project. Their use mainly included improving sentence clarity and grammar, checking LaTeX and Overleaf formatting, helping organise sections of the final paper, and assisting with the presentation of experimental results.

AI tools were also used during parts of the implementation process to help explain programming errors, review code logic, and support the interpretation and organisation of model outputs. Suggestions relating to the comparison of experimental results, structure of the discussion, and presentation of findings were reviewed against the actual implementation outputs and the original research literature before being included in the paper.

The dataset selection, model choices, experimental design, implementation, execution of the experiments, and final academic decisions were carried out by the author with guidance from the project supervisor. All reported results were obtained from the implemented PubMedBERT and Qwen2.5-7B-Instruct pipelines, and citations and technical claims were checked against the relevant original sources.

Responsibility for the accuracy, interpretation, originality, and final content of this paper remains with the author.

Data and Code Availability

The BioRED dataset used in this study is publicly available and is described by Luo et al. [8]. The dataset was obtained from the publicly available BioRED repository.

The source code, preprocessing scripts, model training and evaluation notebooks, and experimental outputs developed for this project are available in the project's GitHub repository at: https://github.com/Mridul33/capstone-project

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